



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2007
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #2

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 61

- Attempt questions 1 - 3

Examiner: *A Fuller*

Total marks – 61
Attempt all questions
All questions are NOT of equal value

Answer each SECTION in a SEPARATE writing booklet

Marks

Section A

Question 1 (19 marks)

- | | | |
|-------|---|----------|
| (a) | Differentiate $\sin^{-1} 3x$ | 1 |
| (b) | Find the exact value of the following: | |
| (I) | $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ | 1 |
| (II) | $\tan^{-1}(-1)$ | 1 |
| (III) | $\sin^{-1}\left(\cos^2 \frac{\pi}{4}\right)$ | 1 |
| (c) | For the function $f(x) = 2 \cos^{-1}\left(\frac{x}{2}\right)$ | |
| (I) | Write down the domain and range | 2 |
| (II) | Sketch the graph of $y = f(x)$ | 1 |
| (III) | Find $f'(x)$ | 1 |
| (d) | Find the indefinite integral of $\frac{1}{\sqrt{1-4x^2}}$ | 2 |
| (e) | ABCD is a cyclic quadrilateral in which the opposite sides AB and DC are equal. | |
| (I) | Draw a diagram | |
| (II) | Prove that the diagonals AC and BD are equal | 3 |
| (f) | Find the value of k if $\int_0^1 \frac{dx}{x^2 + 3} = k\pi$ | 2 |

- (g) A committee of 4 is to be formed from a group of 10 people containing 6 men and 4 women
- (I) In how many ways can this committee be formed so that it always contains at least one woman? 2
- (II) If the selection is determined by the drawing of names from a hat, what is the probability that the committee will contain a majority of women? 2

End of Section A

Section B (Use a SEPARATE writing booklet)

Question 2 (21 marks)

- (a) Use the substitution $u = x^2 - 4$ to find an expression for $\int \frac{2x}{\sqrt{x^2 - 4}} dx$ 2
- (b) An urn contains 3 red, 4 white and 5 blue marbles. Three marbles are drawn at random in succession, and after each marble is drawn it is replaced in the urn. What is the probability that:
- (I) marbles of different colours are drawn? 1
- (II) red, white, and blue marbles are drawn in that order? 1
- (III) at least 2 red marbles are drawn? 1
- (c) Use the method of mathematical induction to prove that 3
- $$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$
- (d) Find the exact area between the curve $y = \sin^{-1} x$, the x -axis, and the 2
- lines $x = 0$ and $x = \frac{1}{2}$
- (e) Find $\cos \left[\sin^{-1} \left(\frac{\sqrt{3}}{2} \right) + \cos^{-1} \left(\frac{\sqrt{3}}{2} \right) \right]$ 2
- (f) Use Newton's method once to find an approximation for $\sqrt{4.02}$ which 2
- is better than 2

- (g) (I) How many five digit numbers can be formed from the digits 1, 3, 5, 7, 9
- (α) without repeating any digit? 1
- (β) allowing each digit to be used any number of times? 1
- (II) Consider now the numbers less than 5000 which can be formed from four or fewer of the digits 1, 3, 5, 7, 9 (without repeats)
- (α) How many of these are there? 1
- (β) One of these numbers is selected at random. What is the probability that it is divisible by 5? 1
- (h) Prove by induction that $2^{3n} - 3^n$ is divisible by 5, if n is a positive integer 3

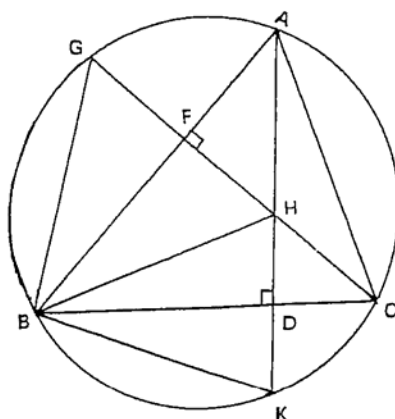
End of Section B

Section C (Use a SEPARATE writing booklet)

Question 3 (21 marks)

- (a) Find the volume formed when the area between the curve $y = \sin x \cos x$, the x -axis, and the lines $x = 0$ and $x = \frac{\pi}{3}$ is rotated about the x -axis 3
- (b) Find the value of $\int_0^{\frac{\pi}{2}} \frac{\sec^2(\sin x)}{\sec x} dx$ using the substitution $u = \sin x$ 2
- (c) Given that $y = \sin^{-1} \sqrt{x}$ show that $\frac{dy}{dx} = \frac{1}{\sin 2y}$ 2
- (d) State the largest domain and the corresponding range of the function $y = \sin^{-1}(\sqrt{1-x^2})$ 2
- (e) (I) Find $\frac{dy}{dx}$ for $y = \cos^{-1}(\sin x)$ 2
- (II) Hence, or otherwise, sketch $y = \cos^{-1}(\sin x)$ for $-2\pi \leq x \leq 2\pi$ 2

(f)



In the above diagram, ABC is a triangle inscribed in a circle.

The perpendicular from A onto BC meets it at D and is then produced to meet the circumference at K.

The perpendicular from C onto AB meets it at F and is then produced to meet the circumference at G.

The two perpendiculars AD and CF meet at the point H.

- | | | |
|-------|--|---|
| (I) | Show that the quadrilaterals AFDC and BFHD are both cyclic | 1 |
| (II) | Prove that AB bisects the angle GBH | 2 |
| (III) | Prove that GB = BK | 2 |

- (g) Given that $\sin x + \cos x = 2^{\frac{1}{2}} \sin\left(x + \frac{\pi}{4}\right)$ and $y = e^x \sin x$, prove by 3

mathematical induction, that $\frac{d^n y}{dx^n} = 2^{\frac{n}{2}} e^x \sin\left(x + \frac{n\pi}{4}\right)$

where $\frac{d^n y}{dx^n}$ denotes the n^{th} derivative of y with respect to x

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

3 unit Section A 2007 May Assess Task 3. (19)

(1) (a) $\frac{d}{dx} (\sin^{-1} 3x) = \frac{1}{\sqrt{1-9x^2}} \times 3 = \frac{3}{\sqrt{1-9x^2}}$ (1)

(b) (i) $\cos^{-1}(\frac{\sqrt{3}}{2}) = \frac{\pi}{6} (30^\circ)$ (1)

(ii) $\tan^{-1}(-1) = -\tan^{-1}1 = -\frac{\pi}{4} (-45^\circ)$ (1)

(iii) $\sin^{-1}(\cos^2 \frac{\pi}{4}) = \sin^{-1}(\frac{1}{2}) = \frac{\pi}{6} (30^\circ)$ (1)

(c) (i) $f(x) = 2\cos^{-1}(\frac{x}{2})$

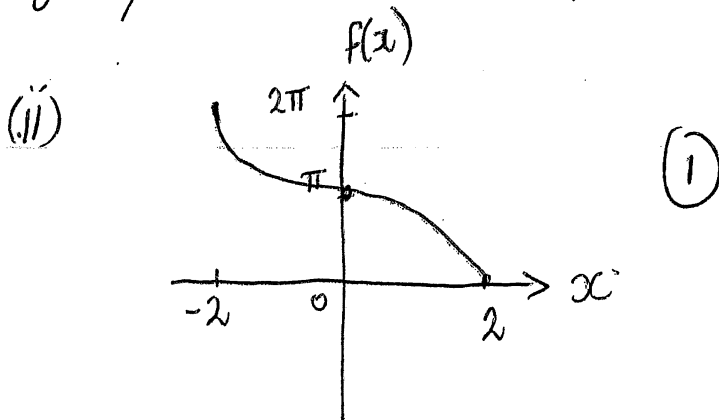
domain of $f(x) = \cos^{-1}x$: $-1 \leq x \leq 1$

domain of $f(x) = 2\cos^{-1}(\frac{x}{2})$: $-1 \leq \frac{x}{2} \leq 1$

$-2 \leq x \leq 2$ (1)

range of $f(x) = \cos^{-1}x$: $0 \leq y \leq \pi$

range of $f(x) = 2\cos^{-1}(\frac{x}{2})$: $0 \leq 2\cos^{-1}(\frac{x}{2}) \leq 2\pi$ (1)



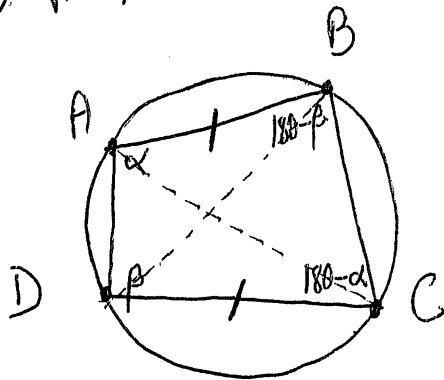
(iii) $f'(x) = 2 \times \frac{-1}{\sqrt{1-\frac{x^2}{4}}} \times \frac{1}{2}$

$= \frac{-1}{\sqrt{4-x^2}} = \frac{-1}{\sqrt{4-x^2}} = \frac{-2}{\sqrt{4-x^2}}$ (1)

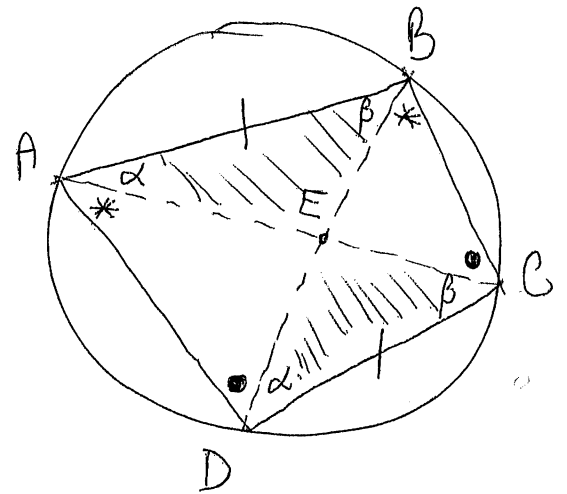
$$(d) \quad \frac{1}{2} \int \frac{21}{\sqrt{1-4x^2}} dx =$$

$$\frac{1}{2} \sin^{-1} 2x + C \quad (2)$$

(e) (i)



$\Rightarrow$



$\hat{A}E = \hat{B}D = \alpha$
angles standing on same arc.
 $\hat{A}BE = \hat{D}CE = \beta$
angles standing on same arc.

$\triangle ABE \equiv \triangle DCE$ (AAS)
 $\therefore \begin{cases} AE = DE \\ BE = CE \end{cases}$ corresponding sides
in congruent triangles

$$AC = AE + CE$$

$$DB = DE + BE$$

$$\therefore AC = DB \text{ as needed.} \quad (3)$$

$$(f) \quad \int_0^1 \frac{1}{x^2+3} dx = k\pi$$

$$\left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} \right]_0^1 = \frac{1}{\sqrt{3}} \left(\tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right)$$

$$= \frac{1}{\sqrt{3}} \cdot \frac{\pi}{6}$$

$$\text{So } k\pi = \frac{\pi}{6\sqrt{3}}$$

$$k = \left(\frac{1}{6\sqrt{3}} \right) \text{ or } \frac{1}{6\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{3}}{18} \quad (2)$$

(9) 10 people: 6 men, 4 women.
Committee of 4.

(1) at least 1 woman.

1 woman + 2 women + 3 women + 4 women
3 men 2 men 1 man

$${}^4C_1 \times {}^6C_3 + {}^4C_2 \times {}^6C_2 + {}^4C_3 \times {}^6C_1 + {}^4C_4$$

$$= 4 \times \frac{6 \times 5 \times 4}{3 \times 2 \times 1} + \frac{4 \times 3}{2 \times 1} \times \frac{6 \times 5}{2 \times 1} + \frac{4 \times 3 \times 2}{3 \times 2 \times 1} \times 6 + 1$$

$$= 80 + 90 + 24 + 1 = 195 \text{ ways}$$

(ii) sample space is ${}^{10}C_4 = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210$

majority of women? 3 women + 4 women = 25
1 man

$$\frac{25}{210} = \frac{5}{42} \text{ (2)}$$

Q2 (a) $u = x^2 - 4$

$$\frac{du}{dx} = 2x$$

$$du = 2x \cdot dx$$

$$\begin{aligned} \therefore \int \frac{2x \, dx}{x^2 - 4} &= \int \frac{du}{\sqrt{u}} \\ &= \int u^{-\frac{1}{2}} du \\ &= 2u^{\frac{1}{2}} + C \\ &= 2\sqrt{x^2 - 4} + C \end{aligned}$$

(b) I $P(RWB) = \frac{3}{12} \times \frac{4}{12} \times \frac{5}{12}$
 $= \frac{5}{144}$

But R, W & B can be arranged in 6 orders

$$\therefore P(3 \text{ different colors}) = \frac{5}{144} \times 6 = \frac{5}{24}$$

II $P(RWB) = \frac{5}{144}$ from I

III $P(E) = P(RRR) + P(RRO) + P(ROR) + P(ORR)$
 $= \frac{3}{12} \times \frac{3}{12} \times \frac{3}{12} + 3 \times \left(\frac{3}{12} \times \frac{3}{12} \times \frac{6}{12} \right)$
 $= \frac{1}{64} + 3 \times \frac{3}{64}$
 $= \frac{10}{64}$
 $= \frac{5}{32}$

(c) $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$

Step 1: Consider the statement for $n=1$

$$\text{LHS} = \frac{1}{1 \times 3} = \frac{1}{3}$$

$$\text{RHS} = \frac{1}{2 \times 1 + 1} = \frac{1}{3} = \text{LHS}$$

$\therefore$ The statement is true for $n=1$

Step 2: Assume the statement is true for $n=k$.

$$\text{i.e. } \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2k-1)(2k+1)} = \frac{k}{2k+1}$$

Consider the statement for $n=k+1$

$$\text{i.e. } \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2k+1)(2k+3)}$$

$$= \frac{k+1}{2(k+1)+1}$$

$$\text{LHS} = \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2k+1)(2k+3)}$$

$$= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \quad \text{using assumption}$$

$$= \frac{k(2k+3) + 1}{(2k+1)(2k+3)}$$

$$= \frac{2k^2 + 3k + 1}{(2k+1)(2k+3)}$$

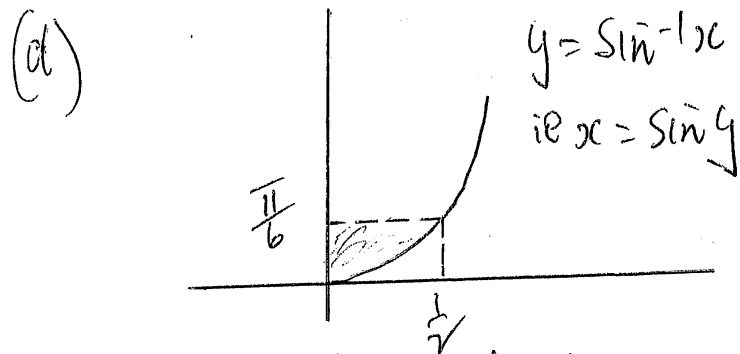
$$= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)}$$

$$= \frac{k+1}{2k+3}$$

$$= \text{RHS}$$

$\therefore$ The statement is true for $n=k+1$ if it is true for $n=k$

Step 5: Now the statement is true for $n=k+1$ if it is true for $n=k$
 But it is true for $n=1$
 $\therefore$ It is true for $n=1+1=2$
 Then it is true for $n=2$, true for $n=2+1=3$
 and so on for all integral n .



Required area = Area rectangle

$$- \int_0^{\pi/6} \sin y \, dy$$

$$= \frac{1}{2} \times \frac{\pi}{6} - [-\cos y]_0^{\pi/6}$$

$$= \frac{\pi}{12} - \left[-\frac{\sqrt{3}}{2} - (-1)\right]$$

$$= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1.$$

(e) Let $\cos\left[\sin^{-1}\frac{\sqrt{3}}{2} + \cos^{-1}\frac{\sqrt{3}}{2}\right] = \cos[A+B]$
 then $\sin^{-1}\frac{\sqrt{3}}{2} = A$ and $\cos^{-1}\frac{\sqrt{3}}{2} = B$
 $\therefore A = \frac{\pi}{3}$ $B = \frac{\pi}{6}$
 $\therefore \cos(A+B) = \cos\left(\frac{\pi}{3} + \frac{\pi}{6}\right)$
 $= \cos\frac{\pi}{2}$
 $= 0.$

(f) $\sqrt{4.2}$ is given by the x intercept
 of $y = x^2 - 4.02$
 ie $f(x) = x^2 - 4.02$ and $f'(x) = 2x$
 Now if $x_1 = 2$ is an approximation to
 a root a better approximation is

given by $x_2 = 2 - \frac{f(2)}{f'(2)}$

$$= 2 - \frac{4 - 4.02}{4}$$

$$= 2 - \frac{-0.02}{4}$$

$$= 2.005$$

(g) I (A) nks = $5! = 120$
 (B) nks = $5^5 = 3125$

II (A) 4 digits $\boxed{2432} = 48$
 3 digits $\boxed{543} = 60$
 2 digits $\boxed{54} = 20$
 1 digit $\boxed{5} = 5$
 Total = 133

(B) 4 digits $\boxed{2321} = 12$
 3 digits $\boxed{431} = 12$
 2 digits $\boxed{141} = 4$
 1 digit $\boxed{1} = 1$
 Total = 29

$\therefore P(E) = \frac{29}{133}$

(h) Step 1: Consider $n=1$

$2^3 - 3 = 8 - 3 = 5$

$\therefore$ True for $n=1$

Step 2: Assume $2^{3k} - 3^k = 5A$

Consider $n=k+1$

$$2^{3k+3} - 3^{k+1} = 2^{3k} \cdot 2^3 - 3^k \cdot 3^1$$

$$= 8 \cdot 2^{3k} - 3 \cdot 3^k$$

$$= 5 \cdot 2^{3k} + 3 \cdot 2^{3k} - 3 \cdot 3^k$$

$$= 5 \cdot 2^{3k} + 3(2^{3k} - 3^k)$$

$$= 5 \cdot 2^{3k} + 3 \cdot 5A$$

$$= 5(2^{3k} + 3A)$$

$\therefore$ Statement is true for $n=k+1$ if it is true for $n=k$ (etc)

SECTION C

QUESTION 3 (3)

(a) $V = \pi \int_0^{\frac{\pi}{3}} (\sin x \cdot \cos x)^2 dx$

$$= \pi \int_0^{\frac{\pi}{3}} \left(\frac{1}{2} \sin 2x\right)^2 dx$$

$$= \frac{\pi}{4} \int_0^{\frac{\pi}{3}} \sin^2 2x dx$$

$$= \frac{\pi}{4} \int_0^{\frac{\pi}{3}} \frac{1}{2} (1 - \cos 4x) dx$$

$$= \frac{\pi}{8} \left[x - \frac{1}{4} \sin 4x \right]_0^{\frac{\pi}{3}}$$

$$= \frac{\pi}{8} \left[\left(\frac{\pi}{3} - \frac{1}{4} \sin \frac{4\pi}{3}\right) - 0 \right]$$

$$= \frac{\pi}{8} \left[\frac{\pi}{3} + \frac{\sqrt{3}}{8} \right]$$

1) $\int_0^{\frac{\pi}{2}} \frac{\sec^2(\sin x)}{\sec x} dx$ (2) let $u = \sin x$
 $du = \cos x dx$
 $x=0, u=0$
 $x=\frac{\pi}{2}, u=1$

$$= \int_0^{\frac{\pi}{2}} \sec^2(\sin x) \cos x dx$$

$$= \int_0^1 \sec^2 u du$$

$$= \left[\tan u \right]_0^1 = \tan 1$$

(c) $y = \sin^{-1} \sqrt{x} \Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}}$ (1)
 $\boxed{x = \sin^2 y}$

(2) $\frac{dy}{dx} = \frac{1}{2 \sqrt{\sin^2 y} \cdot \sqrt{1 - \sin^2 y}}$ (1/2)
 $= \frac{1}{2 \sin y \cdot \cos y}$
 $\frac{dy}{dx} = \frac{1}{\sin 2y}$ (1/2)

(d) (2) $y = \sin^{-1}(\sqrt{1-x^2})$

Domain: $-1 \leq \sqrt{1-x^2} \leq 1$
 $\Rightarrow -1 \leq x \leq 1$

Range: $0 \leq y \leq \frac{\pi}{2}$

(e)

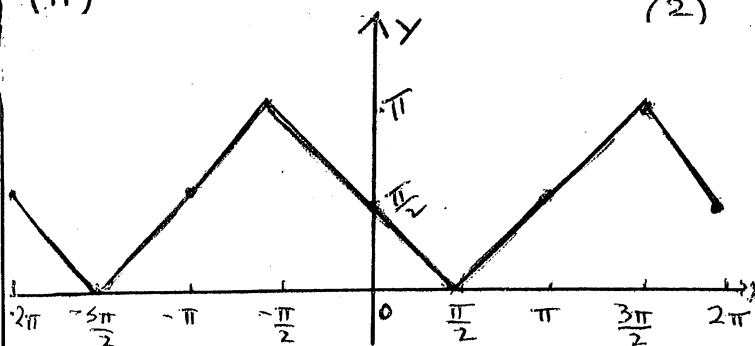
(1) $y = \cos^{-1}(\sin x)$ (2)

$$\frac{dy}{dx} = \frac{-1}{\sqrt{1 - \sin^2 x}} \cdot \cos x$$

$$= \frac{-\cos x}{\sqrt{\cos^2 x}} = \frac{-\cos x}{|\cos x|}$$

$\left\{ \begin{array}{l} \text{If } \cos x < 0, \frac{dy}{dx} = \frac{-\cos x}{-\cos x} = 1 \\ \text{If } \cos x > 0, \frac{dy}{dx} = \frac{-\cos x}{\cos x} = -1 \end{array} \right\}$

(II) (2)



(f) $\hat{AFD} = \hat{ADC} = 90^\circ$

Both angles subtended at the circ. by chord (diameter) AC.
 $\therefore AFDC$ cyclic

also

$\hat{BFH} = \hat{BDH} = 90^\circ$ (opp. angles suppl.)
 $\Rightarrow BDHF$ cyclic

$\hat{GBA} = \hat{GCA}$ (angles at circ. subtended by arc AG)

$\hat{GCA} = \hat{FDA}$ (angles at circ. subtended by chord AF)

$\hat{FDA} = \hat{FBH}$ (angles at circ. sub. by chord FH)

$BDHF$ cyclic quad.

$\Rightarrow \hat{GBA} = \hat{FBH}$

$\therefore AB$ bisects $\hat{GBH}$

$FDCA$ is cyclic quad.

chord FD subtends $\hat{FAD}$ and $\hat{FCD}$ at circ.

$\therefore \hat{FAD} = \hat{FCD}$

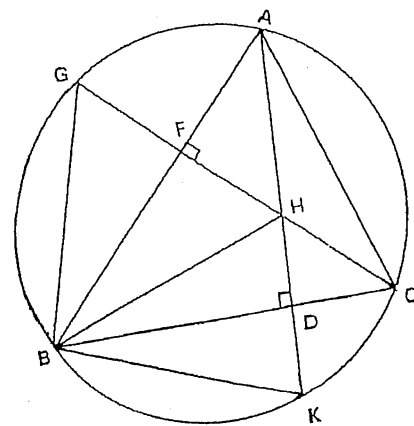
$\Rightarrow$ chord BK subtends $\hat{BAK} = \hat{FAD}$

also chord BG subtends $\hat{ACB}$ at circ. which equals $\hat{FCD}$

$\therefore$ chord $BA =$ chord BK

Since they subtend equal angles at the cir. of large C.

There are other ways of answering these questions.



$$(g) \sin x + \cos x = 2^{\frac{1}{2}} \cdot \sin\left(x + \frac{\pi}{4}\right)$$

$$y = e^x \sin x$$

3) Prove $\frac{d^n y}{dx^n} = 2^{\frac{n}{2}} e^x \sin\left(x + \frac{n\pi}{4}\right)$ — (*) $\frac{dy}{dx} = e^x [\sin x + \cos x] = e^x \cdot [2^{\frac{1}{2}} \sin(x + \frac{\pi}{4})]$

$$\text{When } n=1 \quad \frac{dy}{dx} = 2^{\frac{1}{2}} e^x \sin\left(x + \frac{\pi}{4}\right)$$

$$\text{and } \frac{dy}{dx} = 2^{\frac{1}{2}} e^x \sin\left(x + \frac{1\pi}{4}\right) \text{ using (*)}$$

$\therefore (*)$ true for $n=1$

$$\text{Assume } \frac{d^k y}{dx^k} = 2^{\frac{k}{2}} e^x \sin\left(x + \frac{k\pi}{4}\right)$$

$$\text{RTP } \frac{d^{k+1} y}{dx^{k+1}} = 2^{\frac{k+1}{2}} e^x \sin\left[x + \frac{(k+1)\pi}{4}\right]$$

$$\text{LHS} = \frac{d}{dx} \left(\frac{d^k y}{dx^k} \right)$$

$$= \frac{d}{dx} \left[2^{\frac{k}{2}} e^x \sin\left(x + \frac{k\pi}{4}\right) \right]$$

$$\text{by prod. rule } \Rightarrow = 2^{\frac{k}{2}} \cdot e^x \left[\cos\left(x + \frac{k\pi}{4}\right) + \sin\left(x + \frac{k\pi}{4}\right) \right]$$

$$= 2^{\frac{k}{2}} e^x \left[2^{\frac{1}{2}} \sin\left(x + \frac{k\pi}{4} + \frac{\pi}{4}\right) \right]$$

$$= 2^{\frac{k+1}{2}} e^x \sin\left[x + \frac{(k+1)\pi}{4}\right]$$

$$= \text{RHS}$$